

Sprint Planning Deliverables

Multi-Agent AI Research Assistant

Date: 06 October 2026

Sprint Duration: 10 Working Days

Project Type: AI / Multi-Agent System

This document contains the three requested sprint-planning deliverables: the Sprint Planning Document, Task Priority Table, and Project Timeline.

1. Sprint Planning Document

Project: Multi-Agent AI Research Assistant

Sprint Goal: Develop a functional AI Research Assistant that uses multiple specialized AI agents to research, analyze, verify, and generate a final research report.

Project Overview

The Multi-Agent AI Research Assistant is an AI-based system designed to automate the research process. Instead of using a single AI agent for every task, the system divides the research process among specialized agents.

Agent	Responsibility
Coordinator Agent	Manages and controls the complete workflow.
Research Agent	Collects relevant information for the requested topic.
Analyzer Agent	Analyzes and organizes collected information.
Critic Agent	Reviews information for errors, gaps, and inconsistencies.
Writer Agent	Generates the final structured research report.

Sprint Objectives

- Define project requirements and objectives.
- Design the multi-agent system architecture.
- Set up the development environment and LLM integration.
- Develop the Research, Analyzer, Critic, Writer, and Coordinator agents.
- Implement agent communication and state management.
- Add error handling and validation.
- Test individual agents and the complete workflow.
- Prepare technical documentation and deployment readiness.

Sprint Strategy

The sprint follows a dependency-first approach. Requirements and architecture are completed before development. Core AI functionality is prioritized before testing, documentation, and deployment preparation.

2. Task Priority Table

ID	Project Task	Complexity	Priority	Dependency
T1	Define project requirements	Low	P0 – Critical	None
T2	Design system architecture	Medium	P0 – Critical	T1
T3	Set up Python project	Low	P0 – Critical	T1
T4	Configure LLM/API integration	Medium	P0 – Critical	T3
T5	Develop Research Agent	High	P1 – High	T2, T4
T6	Develop Analyzer Agent	Medium	P1 – High	T2, T4
T7	Develop Critic Agent	Medium	P1 – High	T2, T4
T8	Develop Writer Agent	Medium	P1 – High	T2, T4
T9	Develop Coordinator Agent	High	P1 – High	T5–T8
T10	Implement agent communication	High	P1 – High	T9
T11	Implement state management	Medium	P1 – High	T10
T12	Implement error handling	Medium	P2 – Medium	T10
T13	Test individual agents	Medium	P2 – Medium	T5–T8
T14	Test complete workflow	High	P2 – Medium	T9–T13
T15	Prepare project documentation	Medium	P2 – Medium	T14
T16	Prepare deployment checklist	Low	P3 – Low	T14, T15

Priority Definitions

Priority	Meaning
P0 – Critical	Must be completed first because other tasks depend on them.
P1 – High	Core functionality that provides the main business value.
P2 – Medium	Important for reliability, quality, and maintainability.
P3 – Low	Final preparation activities that are not required for the initial MVP.

3. Project Timeline

The project is planned as a 10-working-day sprint. Core development is completed first, followed by integration, testing, documentation, and deployment readiness.

Day	Planned Activities	Expected Output
Day 1	Requirements analysis and project setup	Requirements document and project structure
Day 2	System architecture and LLM configuration	Architecture design and working LLM connection
Day 3	Research Agent development	Functional Research Agent
Day 4	Analyzer Agent and Critic Agent development	Functional Analyzer and Critic Agents
Day 5	Writer Agent development	Functional Writer Agent
Day 6	Coordinator Agent development	Functional Coordinator Agent
Day 7	Agent communication and state management	Integrated multi-agent workflow
Day 8	Error handling and individual testing	Validated individual components
Day 9	End-to-end workflow testing	Working complete research pipeline
Day 10	Documentation and deployment checklist	Final documentation and readiness report

Sprint Milestones

Milestone	Target	Expected Result
M1	Day 1	Requirements completed
M2	Day 2	Architecture and LLM setup completed
M3	Day 5	Individual agents completed
M4	Day 7	Multi-agent workflow integrated
M5	Day 9	End-to-end testing completed
M6	Day 10	Documentation completed
Final	Day 10	Functional AI Research Assistant MVP

Final Sprint Outcome

At the end of the sprint, the project should provide a functional Multi-Agent AI Research Assistant capable of accepting a research topic, coordinating specialized AI agents, analyzing and validating information, and producing a structured final research report.